

Das Sanjeevan

Native AI Senior Software Engineer | Technical Lead | Java, Spring Boot & Microservices | Generative AI & Agentic Systems
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PROFESSIONAL SUMMARY

Senior Software Engineer and Technical Lead with **6.5+ years of experience** delivering enterprise-scale, high-availability backend systems at Tata Consultancy Services (TCS), serving **millions of users** across large-scale enterprise platforms. Skilled in designing scalable **Java** and **Spring Boot microservices**, **distributed systems**, **RESTful APIs**, and cloud-native platforms that support **mission-critical operations**. Proven record of **leading engineering teams**, engaging directly with enterprise clients, and driving delivery from architecture through production. Currently building and running **Generative AI** and **Agentic AI** solutions in production with **LangChain**, **LangGraph**, **RAG**, **MCP**, and **LLMs** to automate business workflows and improve engineering productivity.

PROFESSIONAL EXPERIENCE

Senior Software Engineer / Technical Team Lead

Tata Consultancy Services (TCS) — Client: CelcomDigi Berhad (Telecom)

Jan 2024 – Present

Kuala Lumpur, Malaysia

Sales Partner Portal (Current)

- Lead the design, development, and enhancement of enterprise-scale backend applications powering CelcomDigi's **Sales Partner Portal**, enabling **dealer onboarding**, partner lifecycle management, and telecom business operations.
- Design and build scalable **Spring Boot** microservices and **RESTful APIs** using **Java**, **Spring Cloud**, **Hibernate**, **JPA**, **Redis**, **MySQL**, and **MariaDB** within a distributed ecosystem of **100+ microservices**.
- Implemented **Elasticsearch**-backed search and retrieval for high-volume partner and transaction data, **reducing query latency** and offloading expensive read patterns from the relational database.
- Secure enterprise APIs and microservices using **Spring Security**, **OAuth 2.0**, **JWT**-based authentication, and **role-based access control (RBAC)** across the platform.
- Built an end-to-end automated test suite using **Playwright** that runs headlessly via a single command — self-executing test scenarios by raising live orders, capturing screenshots of each sensitive screen, and auto-generating and sharing an error report on failure, significantly **reducing manual QA effort** and improving release confidence.
- Collaborate directly with business stakeholders, solution architects, and client leadership to analyse requirements, define solutions, estimate effort, and deliver business-critical features.
- Drive the full software delivery lifecycle — architecture, solution design, development, **code reviews**, testing, **CI/CD** deployment, production releases, and operational support.
- Build and maintain **CI/CD** pipelines using **Azure DevOps**, **GitLab**, **GitHub**, **Jenkins**, and **Maven** to enable reliable enterprise deployments; optimise backend services, database interactions, and **API performance** for scalability and stability.
- Perform production incident analysis, **root-cause investigation**, and critical issue resolution while maintaining **high availability** for enterprise telecom platforms.
- Played a key role in the **Celcom-Digi enterprise platform consolidation**, supporting backend optimisation and seamless migration of **~11 million subscribers** following the telecom merger.

Software Engineer — Dealer Management System

Tata Consultancy Services (TCS) — Client: CelcomDigi Berhad (Telecom)

Jun 2021 – Dec 2023

India / Malaysia

- Built the **Dealer Management System** from scratch as a complete product for CelcomDigi's enterprise sales ecosystem, delivering scalable modules for dealer onboarding, outlet and partner management, and approval workflows.
- Designed and developed high-throughput **Spring Boot** microservices handling **~50,000 API calls per day** with **sub-200ms response times**, ensuring reliable performance under sustained production load.
- Implemented secure authentication and authorization flows using **Spring Security**, **OAuth 2.0**, and **JWT** tokens to protect partner-facing APIs and enterprise user data.
- Owned end-to-end product delivery in a **Scrum** environment — daily stand-ups, weekly sprints, and weekly client demo calls to showcase newly built features and gather feedback.
- Collaborated directly with business users across **Agile** cycles including sprint planning, requirement workshops, **UAT support**, and production deployments, driving the product to a successful **go-live in July 2022**.
- Relocated on-site to **Malaysia** to collaborate directly with client stakeholders and lead continuous platform enhancements through delivery.

Software Engineer — IoT Platform

Tata Consultancy Services (TCS) — Client: Vodafone Idea Ltd. (IoT Platform)

Jan 2020 – May 2021

India

- Designed and implemented the complete **Reverse Logistics Module** supporting device replacement, returns, lost-device processing, and lifecycle management for enterprise **IoT** solutions.

- Architected a reusable, **metadata-driven CRUD framework** enabling configurable database operations, significantly reducing repetitive development effort across the platform.
- Developed backend services using **Java, Spring Boot, Hibernate, JDBC, and MySQL**, ensuring maintainable, scalable, and production-ready architecture.
- Delivered multiple enterprise features following **Agile** practices in close collaboration with business teams.

AI & INNOVATION

Enterprise Telecom Order Recovery Agent

Sep 2025 – Present

Generative AI / Agentic Systems

- Designed, developed, and deployed to production an **AI-powered Order Recovery Agent** that automates telecom **BSS** order-recovery workflows, now running against live enterprise order traffic.
- Built intelligent **multi-agent orchestration** using **LangGraph** and **LangChain** for enterprise workflow automation.
- Implemented **Retrieval-Augmented Generation (RAG)** over telecom SOP documentation on a **Pinecone** vector store to identify failed order states and recommend automated recovery actions.
- Leveraged **OpenAI, Anthropic Claude, Pinecone, MCP, and prompt-engineering** techniques to deliver a production-grade enterprise AI solution.

TECHNICAL SKILLS

- **AI-Assisted Development (Vibe Coding):** Claude Code, GitHub Copilot, Codex, Cursor, Kiro — AI-first, interactive development for shipping production-ready features rapidly
- **LLMs (Hands-on):** Claude Opus 4.5 / Sonnet / Haiku, GPT-4o, o3; multimodal routing and second-opinion AI workflows
- **Generative AI & Agentic Frameworks:** LangGraph, LangChain, Multi-Agent Orchestration, Agentic Frameworks, Tool / Function Calling, Prompt Engineering, Streaming LLM Responses, Context Management Pipelines, Chat Memory & Conversation State, MCP, RAG Pipelines
- **Backend:** Java, J2EE, Spring Boot, Spring MVC, Spring Cloud, Hibernate, Spring Data JPA, REST APIs, Microservices Architecture, Distributed Systems
- **Databases (SQL, NoSQL & Vector):** MySQL, MariaDB, Oracle, Elasticsearch, Redis, Pinecone (Vector DB)
- **Cloud & Messaging:** Microsoft Azure, Apache Kafka
- **DevOps:** CI/CD, Jenkins, Maven, GitLab, GitHub, Docker, Swagger, Postman
- **Security:** OAuth 2.0, JWT, Spring Security, Role-Based Access Control (RBAC), API Security
- **Testing, Automation & Quality:** JUnit, Playwright (E2E test automation), SonarQube
- **Architecture:** API Design, System Integration, Performance Optimisation, Enterprise Applications, Production Support
- **Leadership:** Technical Solutioning, Client Consulting, Team Leadership, Code Reviews, Agile / Scrum Delivery, Sprint Planning, Production Releases
- **Languages:** Java, SQL, Python (Intermediate), JavaScript (Familiar)
- **Tools & IDEs:** IntelliJ IDEA, Eclipse, Spring Tool Suite, VS Code

KEY ACHIEVEMENTS

- Contributed to the successful **Celcom-Digi platform consolidation**, enabling backend optimisation and migration for **~11 million subscribers**.
- Progressed from **Software Engineer** to **Technical Team Lead**, owning solution design, engineering delivery, and client engagement.
- Selected for an **onsite assignment in Malaysia** to collaborate directly with CelcomDigi stakeholders and lead enterprise software delivery.
- Built and deployed to production an **AI-powered telecom Order Recovery Agent** using **LangChain, LangGraph, and RAG** to automate order-recovery workflows.

CERTIFICATIONS

- **Microsoft Certified: Azure AI Engineer Associate (AI-102)**
- **Microservices Architecture with Spring Boot & Spring Cloud** — Udemy

EDUCATION

Bachelor of Technology, Computer Science Engineering

2015 – 2019

B. S. Anangpuria Institute of Technology & Management